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# Magnesium Life Cycle in Automotive Industry

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**Abstract**

The transportation sector is responsible for the largest share (29% in 2019) of greenhouse gas emissions (GHG). One of the ways to reduce this share consists in introducing lightweight vehicles in order to diminish fuel consumption, and, in consequence, GHG emissions. Magnesium (Mg) carries considerable potential as a lightweight material to be used in automotive industry, e.g., in gearboxes, front end and IP beams, steering column and driver's airbag housings, steering wheels, seat frames and fuel tank covers. Weight savings resulting from the use of Mg could reach up to 70% of the conventional parts. However, energy consumption and emissions from magnesium primary production are higher than for steel and aluminum. Therefore, enhancing magnesium recycling would be vital for ensuring its sustainable use. This study assesses the life cycle of magnesium extracted from end-of-life products. Using system dynamics modelling, we quantify the potential environmental benefits and examine aspects related to circularity of magnesium to be used in the automotive industry. Energy consumption, water use and related emissions are assessed in processes of functional (recovered Mg reused in the vehicle manufacturing processes) and non-functional (recovered Mg as an element used in aluminum alloys) recycling as well as casting and molding. Obtained results show that the implementation of circular economy strategies may help in increasing the supply of magnesium from the automotive industry up to 0.70 million tonnes (mt) in 2050. As shown by the analysis, 43% of total energy consumption and water use at the global Mg recycling stage in 2050 corresponds to non-functional casting and molding in automotive industry, followed by non-functional recycling (11%), functional casting and molding (7%) and functional recycling (2%). Also, it is estimated that the existing technologies in functional and non-functional Mg recycling in automotive industry contribute to the global warming with an emission of 5.62 and 29.66 mt of $CO_2$ eq. in 2050, respectively.

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*Keywords:* Critical materials; Circular economy; Environmental sustainability; Dynamic modelling

## 1. Introduction

Magnesium (Mg) is the eighth most abundant element available (2.7% of earth's crust). Due to its relatively high supply risk and economic importance magnesium was identified as a critical material by the EU Commission. Top producers of magnesium from primary sources in 2019 are China (970 kilotonnes (kt)), Russia (67 kt), Kazakhstan (25 kt), and Israel (21 kt) [1]. According to the European Commission's report [2], magnesium is used mainly in automotive industry (44%), packaging (19%), construction (12%), desulfurization (11%), transportation including air transport, maritime and railway transport (4%), and other applications (10%). The major use of Mg in the automotive industry, where it is used as the lightest among structural metals (one-third less dense than aluminum (Al)), can be explained by high specific strength making Mg particularly useful for the production of lightweight cars.

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Global demand for Mg has been increasing rapidly, particularly for the die casting sector (6.8% increase) and Al alloys (4.1% increase) [2]. The major reason for this growth is the increasingly severe regulations on emissions from vehicles and rising cost of petrol. Vehicles account on average for 90% of energy consumption in their life cycle during the exploitation phase. The European Parliament and the Council adopted a Regulation on $CO_2$ emissions in the automotive sector (EU Regulation 443/2009) and mandatory driving cycle $CO_2$ emission performance standards for new passenger cars. According to this Regulation, a target of 95 g $CO_2$/km has been set for EU 2020/21 (possibly 10 g $CO_2$/km in 2050) [3]. As a result, car manufacturers need to improve fuel efficiency of vehicles. One solution is to make them lighter and a way to achieve this goal is to use more magnesium [4].

Considering the increasing amounts of magnesium used in car manufacturing as structural metal that reduces vehicle weight, fuel consumption, and greenhouse gas emissions (GHG) [5], recycling end-of-life vehicles (ELVs) to recover Mg becomes a very relevant step in the automotive industry. Also, if secondary ingots and Mg alloys could be used as die cast products for vehicles, similar to aluminum, an additional reduction of $CO_2$ could be possible [6].

Most magnesium is produced in China. Its processing and primary production are very energy-intensive and generate large amounts of GHG emissions, which may offset the potential advantage of using Mg parts in automobiles [7]. Therefore, circular economy of magnesium could significantly contribute to the reduction of energy and water consumption and mitigate energy-related emissions compared to primary production.

The main objective of this paper is to present a dynamic model of the magnesium supply chain with a focus on automotive industry, to perform environmental sustainability assessment for Mg recycling in line with circular economy concept. Therefore, this study attempts to assist in understanding to what extent Mg recycling contributes to its sustainable use from environmental standpoints.

## 2. Materials and Methods

In this study, the United States Geological Survey (USGS) [1], European Commission [2], GREET-2020 software [8], Statista database [9], and other studies [10], [11] are used as the primary data sources.

The system considered in this study focuses on the complex interrelations among several stages such as mining, processing, manufacturing, and recycling to assess the circularity of magnesium. The causal loop diagram (CLD) by using the principles of system dynamics modelling is depicted in Fig. 1. The CLD helps to identify the key variables that affect system and their interactions with each other. Next to the arrowhead, the positive (+) or negative (-) signs show if the linked components change in the same or reverse direction. It can be seen from Fig. 1 that two types of loops exist in this system including balancing (B) and reinforcing (R). The model shows that there are five balancing loops and two reinforcing loops as follows:

- B1 loop includes *Manufactured semi-finished products* and *Manufacturing demand*
- B2 loop includes *Use in transportation sector* and *Market demand*
- B3 loop includes *Manufactured semi-finished products*, *Use in transportation sector*, *Market demand*, and *Manufacturing demand*
- B4 loop includes *Manufactured semi-finished products*, *Use in transportation sector*, *Collection of waste containing Mg*, *Functional recycling*, *Recovery of magnesium*, *Primary magnesium production*, and *Processed magnesium*
- B5 loop includes *Manufactured semi-finished products*, *Use in transportation sector*, *Collection of waste containing Mg*, *Non functional recycling*, *Recovery of magnesium*, *Primary magnesium production*, and *Processed magnesium*
- R1 loop includes *Manufactured semi-finished products*, *Use in transportation sect*or, *Collection of waste containing Mg*, *Functional recycling*, and *Recovery of magnesium*
- R2 loop includes *Manufactured semi-finished products*, *Use in transportation sect*or, *Collection of waste containing Mg*, *Non functional recycling*, and *Recovery of magnesium*

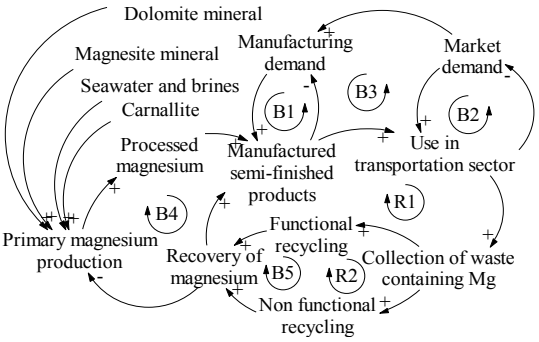



Fig 1. Causal loop diagram of supply chain of magnesium

Magnesium flows are analyzed from the mining stage to track the primary Mg metal in automotive industry. Primary magnesium is mainly extracted from minerals such as dolomite, magnesite and carnallite (respectively 47%, 19% and 13% of magnesium oxide output), as well as seawater and brine (21% of global Mg) [12]. Primary Mg principally used for casting in the automotive industry accounts for 55% of its consumption. Aluminum-based alloys used in the same industry for similar applications account for 28%.

Primary metallic magnesium is produced by using two processes, i.e. thermal reduction and electrolysis [13]. China whose share in the Mg market in 2019 was 85.7% uses thermal reduction method, the so called Pidgeon process. Other large producers of Mg such as Russia (7.0%), Israel (3.9%), Kazakhstan (2.8%), Ukraine (0.3%) use the electrolytic process. The USA having 6.3% share in the Mg market and Canada (1.1%) use both thermal reduction and electrolytic processes [7].

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Table1. Equations of dynamics model for mass flow of magnesium in automotive industry.

| Equations | Description | Type | Unit |
|---|---|---|---|
| $\boldsymbol{Mg_w(t) = \int_{t0}^{t} \left( M_i(t) - N_j(t) \right) dt + Mg_w(t_0)}$ | $Mg_w(t)$ is the cumulative amount of global primary magnesium in the mining stage in year $t$. $t_0$ is an initial time. | Stock | Tonnes |
| $\boldsymbol{M_i(t) = \sum_{i=1}^{2} f_i \times \alpha_i}$ | $M_i(t)$ is the annual rate of extracted Mg. $f_i$ is the rate of processing $i$ ($i$=1,2) including minerals and seawater or brines. $\alpha_i$ is the conversion factor of each process. | Flow | Tonnes |
| $\boldsymbol{N_j(t) = \sum_{n=1}^{2} k_j \times \beta_j}$ | $N_j(t)$ corresponds to the annual rate of magnesium production. $k_j$ is the rate of production $j$ ($j$=1,2) including metallic Mg and Mg alloys. $\beta_j$ is the conversion factor of each process. | Flow | Tonnes |
| $\boldsymbol{Mg_a(t) = \int_{t0}^{t} \left( N_j(t) - Y_i(t) \right) dt + Mg_a(t_0)}$ | $Mg_a(t)$ is the cumulative amount of Mg used in automotive industry in year $t$. $t_0$ is an initial time. | Stock | Tonnes |
| $\boldsymbol{Y_i(t)}$ = Min ($\boldsymbol{Mg_a(t) \times C_i, D_i}$) | $Y_i(t)$ is rate of Mg used for vehicle manufacturing. The mathematical function is considered as MIN (A, B), i.e. minimum of two alternatives. $C_i$ is capacity of using Mg for car manufacturing. The capacity is defined based on the consumption of magnesium as lightweight material in manufacturing of vehicles. $D_i$ represents demand for Mg in automotive industry. Index $i$ corresponds to Mg metal or alloy. | Flow | Tonnes |
| $\boldsymbol{W_a(t)}$ = Delay ($\boldsymbol{Y_i(t) \times e, L}$, 0) | The function formulates as: Delay (Input, Delay Time, Initial Value). $W_a(t)$ is collected waste and scrap including Mg from end-of-life vehicles. $e$ is efficiency of collection of end-of-life vehicles. $L$ is automobile average lifetime (average vehicle age). | Flow | Tonnes |
| $\boldsymbol{Mg_r(t) = \int_{t0}^{t} \left( W_a(t) - (Mg_r(t) \times (\varrho_i + \delta_i + \upsilon_i) \right) dt + Mg_r(t_0)}$ | $Mg_r(t)$ is the cumulative amount of recyclable Mg collected in automotive industry in year $t$. $\varrho_i$ is efficiency of the disposal of Mg from transportation sector. $\delta_i$ is efficiency of functional recycling of Mg in automotive industry. $\upsilon_i$ is efficiency of non-functional recycling of Mg in automotive industry. | Stock | Tonnes |

Secondary magnesium can be obtained from pre- and post-consumer scrap. Pre-consumer scrap arises from manufacturing process in the form of chips and discards or as scrap during the production of components [14]. Post-consumer scrap comes from end-of-life cars after they have been dismantled and their parts were sold in the secondary market.

Table 1 provides the variables developed for calculating the stock and flow of magnesium for automotive industry from mining to recycling. The world production of primary magnesium which their historical data is available by USGS in the period (1990-2018) [1] is considered in order to calculate the uncertainty of the model and its validity. The average percentage error of the simulation model is 1% and considering 95% level of confidence the model is valid.

## 3. Results and discussion

### *3.1 Trends of generation of magnesium waste and scrap*

Standard parts of vehicles made of magnesium alloys can be found in the instrument panel, support frame, seat frame, steering wheel core parts, steering wheel frames, cylinder block heads, transmission cases, clutch housings, lower crank cases, intake manifolds, brake and gas pedals. The main applications of magnesium alloy in car manufacturing are shown in Fig. 2a. Using Mg in car manufacturing in applications presented in

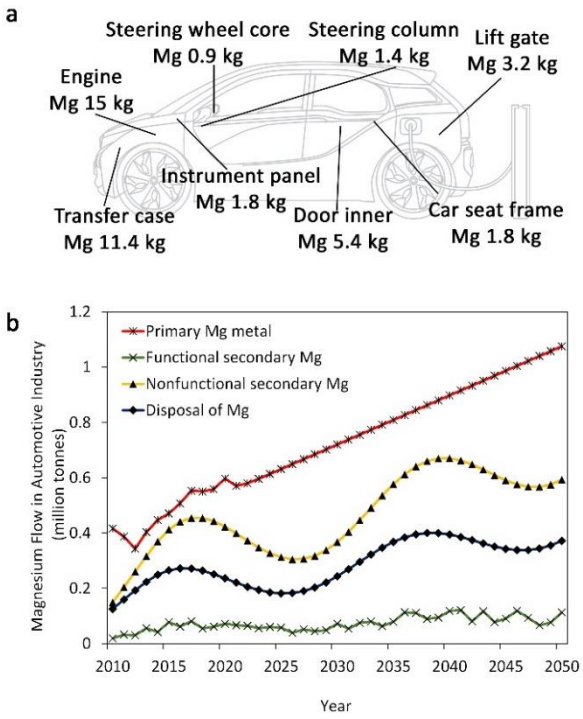



Fig 2. Primary and secondary production of magnesium used in automotive industry from 2010 to 2050. (a) application of Mg in a vehicle; (b) mass flow of primary and secondary magnesium.

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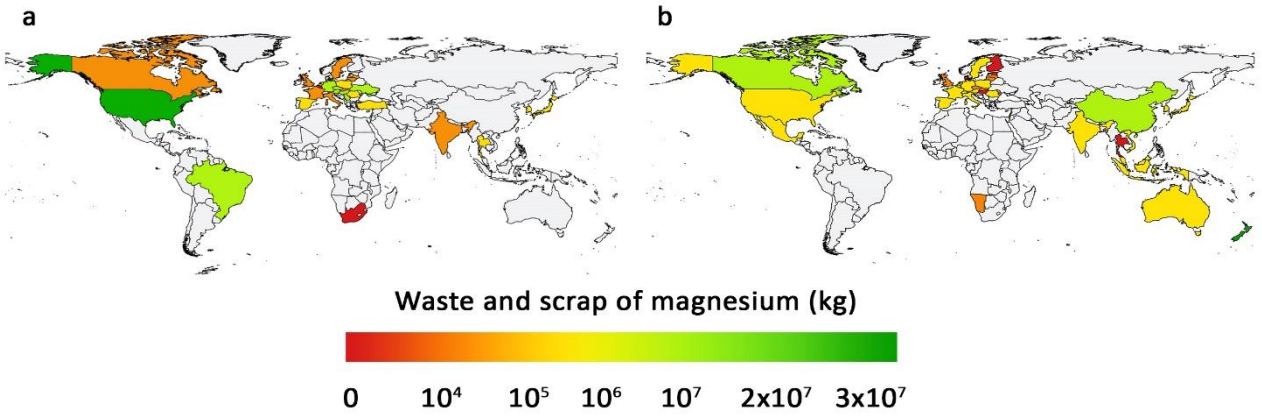



Fig. 3. Global trade of magnesium waste and scrap in 2020. (a) imports; (b) exports.

Fig. 2a allows to achieve 22% to 70% weight reduction of components of a vehicle [15].

Fig. 2b represents flows of primary and secondary (through functional and non-functional recycling) magnesium production consumed by automotive industry and its disposal. Functional secondary Mg corresponds to recovered metal which is reused in the vehicle manufacturing processes, mainly within die casting industry to produce lightweight vehicles. Non-functional secondary Mg corresponds to recovered Mg as an alloy element which is essential for aluminum industry to improve mechanical properties, deformability, and corrosion resistance of Al alloy.

The highest amounts of magnesium recovered from the automotive industry come from non-functional recycled magnesium (55−80%). To adopt the simulation model in the long-term perspective (2010-2050), one of the most important causes which is needed to be considered is the delay mechanisms of the system. As mathematical function of model presented in Table 1, the differences in the flows of non-functional secondary Mg and the disposal of Mg are mainly affected by the lifespan of passenger cars which is on average ~10.8 years. For disposal purposes, Mg is usually collected as dust or sludge and stored in disposal sites or could go through controlled burning. Projected amount of magnesium in 2050 is 0.59 million tonnes (mt), 0.37 mt and 0.11 mt respectively for non-functional secondary Mg, Mg disposal and functional secondary Mg. Also, global amount of primary magnesium used in the automotive industry is estimated to increase from 0.4 mt in 2010 to 1.1 mt in 2050. Noteworthy, primary production of magnesium can be associated with higher levels of waste production when compared to recycling processes. Therefore, secondary production of magnesium contributes directly to the achieving of the targets of the 2030 Agenda for Sustainable Development Goals (SDGs) 12 intended to ensure sustainable consumption and production patterns.

To analyze the availability of Mg for recycling in line with circular economy principles, global trade in magnesium waste and scrap in 2020 is shown in Fig. 3 (data retrieved from https://comtrade.un.org/). Trade in magnesium corresponds to 0.01% of total world trade. The exports of magnesium increased by 4.5%, from USD 1.81 billion to USD 1.89 billion, between 2018 and 2019.

The top three exporters of magnesium waste and scrap in 2020 (Fig. 3a) are New Zealand (26.2 kt in trade worth USD 7.4 million), Canada (9.1 kt in trade worth USD 22.1 million), China (6.6 kt in trade worth USD 14.1 million). The top importers of magnesium waste and scrap in 2020 (Fig. 3b) are the USA (23.8 kt in trade worth USD 55.0 million), the Czech Republic (5.6 kt in trade worth USD 11.5 million), Hungary (3.9 kt in trade worth USD 8.1 million), Germany (3.4 kt in trade worth USD 4.9 million), Austria (2.3 kt in trade worth USD 3.4 million), and Ukraine (1.7 kt in trade worth USD 1.8 million). The analysis shows that the Czech Republic accounted for 27% of all Mg waste and scrap in Europe in 2020. Focusing on the automotive industry, Fig. 4 presents the amount of magnesium waste from vehicles available for recycling in Europe. Data for 30 countries including Austria, Belgium, Bulgaria, Switzerland, Czech Republic, Germany, Denmark, Spain, Finland, France, UK, Greece, Hungary,

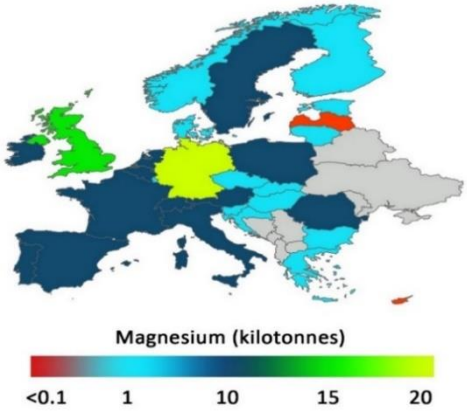



Fig 4. Magnesium waste from automotive industry in Europe in 2020.

Ireland, Italy, Lithuania, Latvia, Norway, Netherlands, Poland, Portugal, Romania, Slovenia, Slovakia, Sweden, Cyprus, Croatia, Luxembourg, Estonia, and Malta are available and analyzed in Europe (data retrieved from https://comtrade.un.org/). Because of the high economic value of the scrap recycling process, especially in Europe, global

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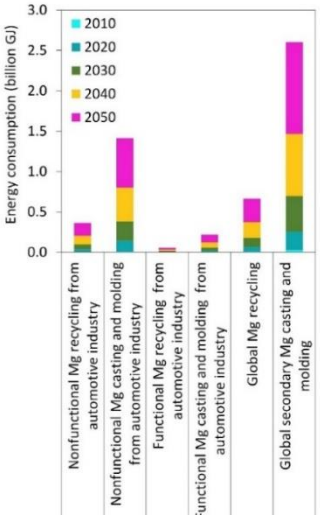



Fig 5. Global energy consumption for secondary Mg production.

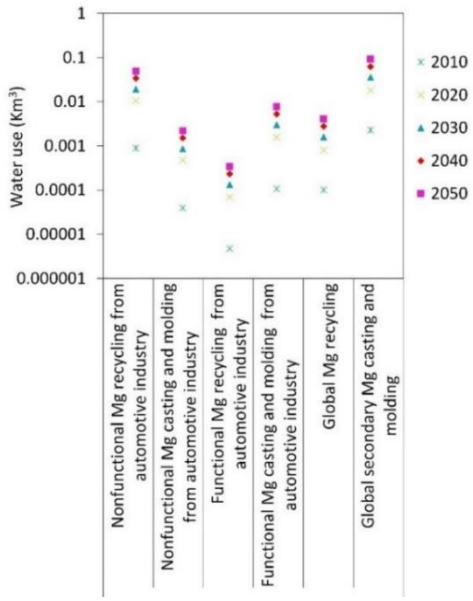



Fig. 6. Global water use for secondary Mg production.

trade imbalances make trade in Mg scrap economically attractive in this area. Top countries with the highest amount of magnesium waste from vehicles are: Germany (16.7 kt), United Kingdom (10.1 kt), and France (9.4 kt). Total magnesium scrap available from the automotive industry in 2020 is estimated around 73.8 kt, however, it should be taken into account that the yield for scrap recycling from the ELVs is approximately 85–95% while the efficiency of car dismantling is up to 90% which depends on the separation procedures and on the composition of the shredder material [16].

### *3.2 Energy consumption and water use for secondary Mg production in automotive industry*

This section presents trends in global energy consumption (Fig. 5) and global water use (Fig. 6) for secondary Mg production focusing on waste and scrap from the automotive industry that consumes more than 50% of all produced magnesium. Two main stages including Mg recycling as well as Mg casting and molding are considered for environmental analysis.

The results indicate an increase in global energy consumption and water use for Mg recycling from 2010 to 2050. The growth rates in the first decade (2010-2020) increased ca. 7 times, 11 times and 14 times for global Mg recycling, non-functional Mg recycling and functional Mg recycling, respectively. This is mainly affected by the amount of waste available from ELVs. It is estimated that total energy consumption and water use for global Mg recycling in 2050 will reach 0.29 billion GJ and 0.004 $Km^3$, respectively. Also, total energy consumption and water use for global casting and molding of secondary Mg will reach 1.14 billion GJ and 0.09 $Km^3$ in 2050.

Non-functional Mg casting and molding in automotive industry corresponds to 43% of total energy consumption and water use for global recycling stage in 2050, followed by non-functional Mg recycling (11%), functional Mg casting and molding (7%) and functional Mg recycling (2%). This shows that secondary production of magnesium is much less energy intensive compared to the primary production. However, our findings suggest that there are still substantial environmental issues involved in recycling activities, especially in the automotive industry, which indicate that the existing technologies need to be revised and improved.

### *3.3 Environmental impact of secondary Mg production*

In this study, eleven different pollutants are analyzed for Mg recycling in automotive industry (Fig. 7). Emissions of carbon dioxide ($CO_2$), methane ($CH_4$), sulfur oxides ($SO_x$), nitrogen oxides ($NO_x$), carbon monoxide (CO), volatile organic compounds (VOC), particulate matter of a diameter smaller than 2.5 micrometer (PM2.5), nitrous oxide ($N_2O$), particulate organic carbon (POC), black carbon (BC), and airborne particulate matter smaller than 10 micrometer (PM10) are assessed for Mg recycling (Fig. 7 a) and Mg casting and molding (Fig. 7 c). Also, we quantify the trend in greenhouse gas emissions of the functional and non-functional Mg recycling (Fig. 7 b) as well as casting and molding (Fig. 7 d) in automotive industry compared to global Mg recycling.

It is estimated that functional and non-functional Mg recycling in automotive industry carried out using the existing technologies contributes to the global warming with emissions of about 5.62 and 29.66 million tonnes of $CO_2$ eq. in 2050. It is worth noting that if for recycling purposes electricity could be generated from renewable energy sources, $CO_2$ emissions from the process could also be reduced [6]. More detailed calculations show that global Mg recycling will generate 1.79 billion tonnes of $CO_2$ eq emissions by 2050.

Casting and molding stage of secondary Mg recycling had impact on global warming with about 0.75 and 2.98 million tonnes of $CO_2$ eq. by 2010 for functional and non-functional molding and casting and Mg recycling. Due to production increases, GHG emissions for the same processes will increase

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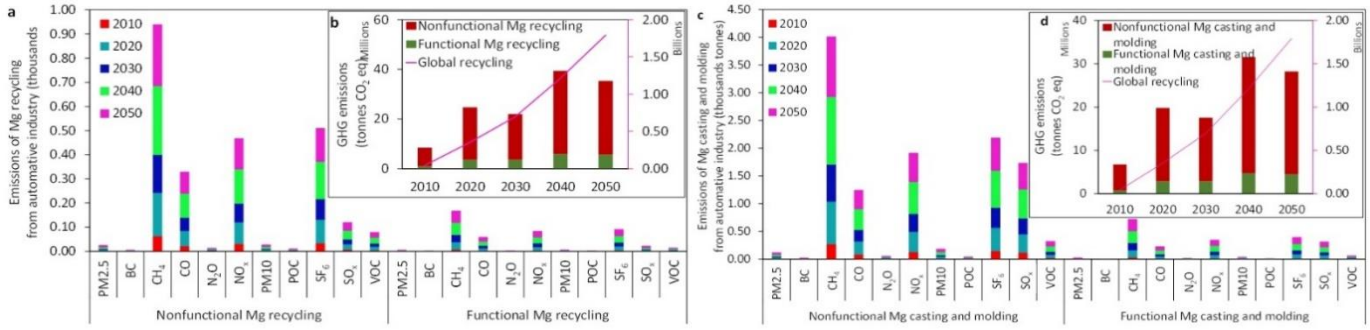



Fig 7. Global emissions of functional, non-functional and global Mg recycling as well as casting and molding in 2010-2050 from automotive industry.

by 6 and 3.97 times and will reach 4.50 and 23.73 million tonnes of $CO_2$ eq. by 2050. Global secondary Mg casting and molding will generate 1.43 billion tonnes of $CO_2$ eq emissions by 2050. The highest amount of emmisions for both Mg recycling as well as casting and molding are reported for $CH_4$, CO, $NO_x$ and $SF_6$, especially in non-functional secondary Mg processes. The finding highlights the necessity of improvement of the current recycling technologies to mitigate the above-mentioned emissions. This finding addresses the targets of SDGs 13 − mitigation of climate change.

## 4. Conclusion

Considering circular economy principles − reusing and recycling rather than discharging materials as waste − we analyzed environmental sustainability of magnesium recycling in the automotive industry. Implementation of circular economy strategies contributes to the recovery of significant amounts of magnesium reaching up to 0.59 mt of non-functional Mg and 0.11 mt of functional Mg in 2050. However, these findings highlight the need to improve magnesium recycling technologies especially in non-functional casting and molding. The result shows that 43% of total energy consumption and water use of global Mg recycling stage in 2050 corresponds to non-functional casting and molding in the automotive industry. Also, we demonstrated that environmental footprint of Mg recycling corresponds mainly to high-level emissions of $CH_4$, CO, $NO_x$ and $SF_6$. It is estimated that the existing technologies in functional and non-functional Mg recycling in the automotive industry contribute to the global warming with emissions of 5.62 and 29.66 mt of $CO_2$ eq. in 2050, respectively. Moreover, the analysis presented in this study shows that the functional Mg recycling is negatively affected by the lack of established value-added chains for end-of-life magnesium scrap in automotive industry.

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